Two-stage Process Analysis Using the Process-based Performance Measurement Framework and Process Simulation

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Abstract

In the diagnosis phase of business process management (BPM), the operational processes are analyzed to identify problems and to find things that can be improved. And in the process design phase of BPM, to-be process is newly designed and analyzed. Proposed in this paper is a two-stage process analysis for process (re)design using the process-based performance measurement framework (PPMF) and process simulation. A two-stage process analysis consists of macro analysis and micro analysis. The most influencing business process on specific strategic level KPI is selected at the macro analysis stage. Afterwards, the performance of selected process is reviewed, and new to-be process is redesigned. The performance prediction of new process is conducted using process simulation at the micro analysis stage.

By using the proposed two-stage process analysis, company practitioners involved in the process innovation can easily and efficiently find the most influencing processes upon enterprise strategy, and evaluate the performance prediction of newly designed process.

1. Introduction

In today's dynamic business environment, the ability to improve business performance is a critical requirement for organization. So many enterprises have recently been pursuing process innovation or improvement to attain their performance goal. A business process is a sequence of activities that carries out a complete business goal. Business process management (BPM) is the identification, understanding and management of business process that link with people and systems in and cross organizations [1].

BPM cycle is composed of diagnosis, process design, system configuration, process enactment [2]. In the diagnosis phase, the operational processes are analyzed to identify problems and to find things that can be improved. And in the process design phase of BPM, to-be process is newly designed and analyzed. According to Van der Aalst et al., there are basically three types of business process analysis: validation, verification and performance analysis [3]. Performance analysis is concerned with evaluating the ability to meet requirements with respect to throughput times, service levels, and resource utilizations.

Performance measurement is needed for performance analysis, and metrics should be established for performance measurement. Since metrics show how well the business is doing relative to a defined strategy, they help managers to derive better business decisions. Because enterprise performance is achieved through the execution of business processes, business process and enterprise performance measurement are closely interrelated. In other words, for process innovation or improvement, the process which most significantly produces a successful performance should be identified, as should the performance indicator which is most significantly influenced when a specific process is executed efficiently. Consequently, performance measurement cannot be managed independently of business processes in order to attain the goal of an enterprise.

Proposed in this paper is a two-stage process analysis for process (re)design using the process-based performance measurement framework and process simulation. A two-stage analysis consists of macro analysis and micro analysis of business process as follows: At the early stage of business process analysis, high level analysis is needed such as the influence of a business process on a specific key performance indicator (KPI) and the contribution of a specific KPI to other KPIs (macro analysis). If certain processes that need to be improved are identified

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through the macro analysis and to-be processes are newly designed, detail level analysis using simulation is needed for each process (micro analysis). Simulation is widely used for effective tool for detailed business process analysis, because it is positioned as a means to evaluate the impact of process changes and new processes in a model environment through the creation of what-if scenarios. Simulation is also promoted to enable examination and testing of decisions prior to actually making them in the real environment [4]. The macro and micro analysis in this paper are based on the well-established process-based performance measurement framework (PPMF) [5].

Of the research related to the performance measurement of an enterprise, Kuwaiti and Kay summarized the role and importance of performance measurement to conduct business process reengineering (BPR) through a survey of many enterprises [6]. Slater et al. classified enterprise strategy into five categories by modifying the four perspectives of balanced scorecard (BSC), and extracted KPIs for each category [7]. However, they did not explore the relationship between business process and performance measurement. Ghalayini et al. classified the performance measures for manufacturing enterprise into the following three layers: general area of success, performance measure, and performance indicator [8]. They defined hierarchical relationships between the layers, but did not show the relationship between performance measure and business process. Berler et al. classified a hospital's KPIs based on the concept of BSC [9]. They also defined hospital workflows in detail, and then assigned a KPI to each unit task-level activity. However, they addressed the interrelationship between KPI and business process merely by simple one-to-one mapping at a single flat level. Chan and Beretta decomposed the business process hierarchically into sub-processes, and then proposed a method of calculating the performance measure by summing up the measures of the lowest level activities [10] [11]. Theirs is a top-down approach that defined the business processes level-by-level, and mapped the performance measure to each business process. Nevertheless, they suggested that the business process has a simple one-to-one relationship with a performance measure. Bititci conducted a case study that applied the information systems modeling techniques to the modeling of a performance measurement system and implied that KPIs are interrelated with multiple and weighted relationships [12]. Munehira et al. suggested that it is necessary to specify which processes should be improved or created to achieve the business performance goals described in BSC [13]. For this purpose, they defined the relations between BSC structure and business model elements, yet they did not implement their proposed method to a real business case.

In the area of business process simulation (BPS), April et al. suggested the effectiveness of simulation for the business process redesign by presenting the case study of a personal claims process at an insurance company [4]. Mevius and Oberweis presented that Petri-net based simulation is useful for the evaluation of process performance [14]. Greasley showed the ability of BPS to incorporate system variability, scenario analysis and a usual display to communicate process performance for BPR [15]. An and Jeng presented the development of system dynamics model of business processes for accurate description of system behavior and what-if analysis [16]. But their model focused on detailed process behavior having single flat level description.

The rest of this paper is organized as follows. Section 2 describes the structure and application of the proposed PPMF. Section 3 presents the method of two-stage process analysis. Finally, the last section presents the conclusions and suggests directions for future research.

2. Process-based performance measurement framework

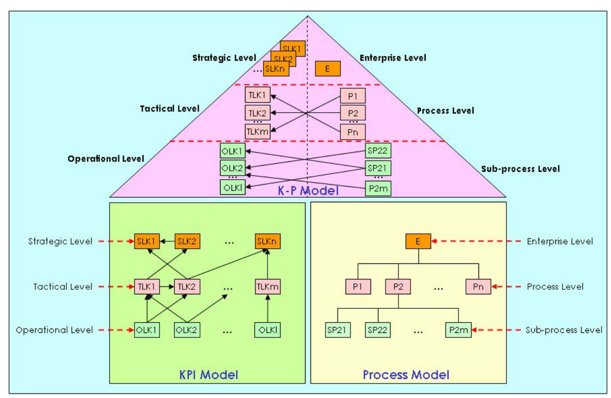

Figure 1. PPMF structure

The need for systematic performance measurement based on business processes has been steadily increasing. However, current measurement practices have not clearly defined and managed the relationships between business processes and KPI. It is therefore difficult to decide which process should be improved to achieve a specific performance goal or which performance index is influenced when a specific business process is executed successfully. Consequently, a performance measurement framework that is closely related to business processes should be established to achieve the goal of an enterprise.

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The proposed PPMF consists of three sub-models, as shown in figure 1: KPI model, Process model, and K-P model. Firstly, KPIs are hierarchically classified into three levels in the KPI model according to the following management decision level: strategic level KPI (SLK), tactical level KPI (TLK), and operational level KPI (OLK). The contribution index, which is a measure of the contribution of a specific KPI to another or higher level KPI, is determined in the KPI model.

Secondly, business processes are classified into three levels in the process model in accordance with the size of the process span as follows: enterprise level, process level, and sub-process level. Each sub-process has a network composed of unit activities as shown in figure 2. Process simulation for micro analysis is conducted at the sub-process level.

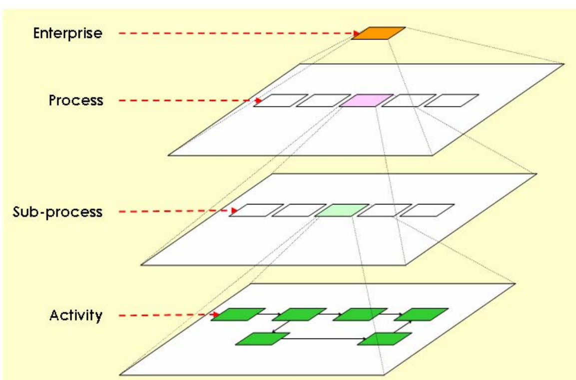

Figure 2. Concept of process model

Finally, the K-P model represents the relation between KPI and business process. Enterprise, process, and sub-process levels in the process model correspond to SLK, TLK, and OLK in the KPI model, respectively. The influence index, which is a measure of the influence of a business process on a specific KPI is determined in the K-P model.

The procedure to develop the PPMF is divided into four phases, as shown in figure 3. Phases I and III are concerned with the construction of the KPI model. The process model is constructed in phase II. Lastly, the K-P model is built in phase IV. During the procedure, the KPI and process models are interleaved because it is difficult to identify KPIs in isolation from business processes.

The following three steps are conducted during phase I. Firstly, SLKs of the enterprise are identified. Secondly, the major business areas of the enterprise are defined. Finally, existing KPIs for each business area are identified and refined to become TLK or OLK candidates.

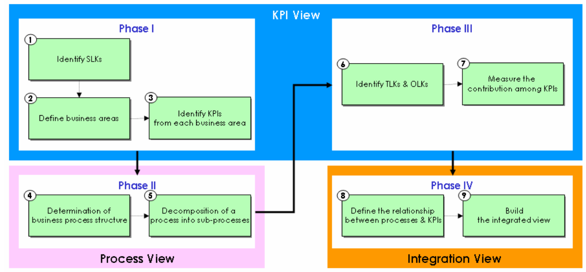

Figure 3. Procedure for PPMF development

Phase II consists of two steps in which the major activities are business process and sub-process definition. Firstly, business process structure of company is determined. Secondly, each business process is decomposed into sub-processes. Figure 4 shows the process model of the case company S. S company is one of the biggest companies in Korea's shipbuilding industry with about 7,000 employees, and mainly builds large ships.

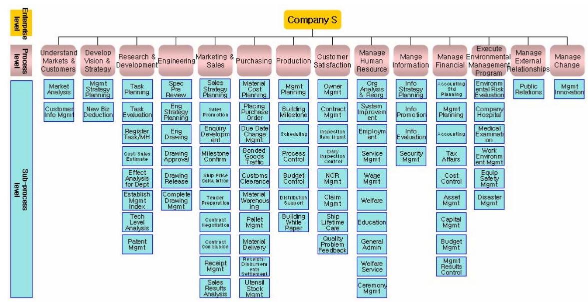

Figure 4. Process model of S company

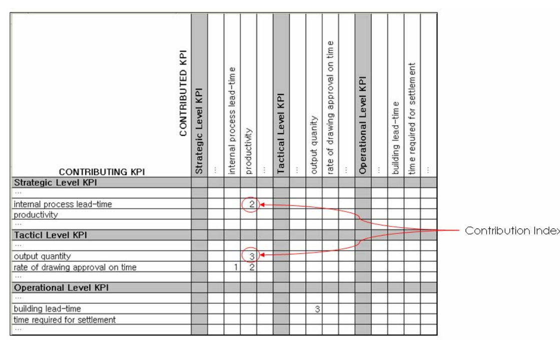

Figure 5. KPI model (simplified)

During phase III, the KPI model is constructed completely based on the results of phase I. Phase III consists of two steps in which TLK and OLK are identified in terms of the business process, and the contribution indexes among KPIs are determined. Contribution indexes are measured for each SLK, TLK, and OLK to the same level and higher level KPIs with a 3-point scale (1: minor contribution, 2: normal contribution, 3: major contribution). Figure 5 shows

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the KPI model of the case company S. In figure 5, the contribution index of ‘output quantity’ TLK to the ‘productivity’ SLK is marked as 3.

Phase IV consists of two steps in which the major activities are the measurement of influences of each process upon KPIs and the building of an integrated view of KPI and processes. The influence indexes are measured for the process on TLKs and the sub-process on OLKs with a 3-point scale (1: minor influence, 2: normal influence, 3: major influence). Figure 6 shows the integrated view of PPMF of the case company S. In the left side of figure 6, influence index of ‘production’ process upon the ‘output quantity’ TLK is marked as 3.

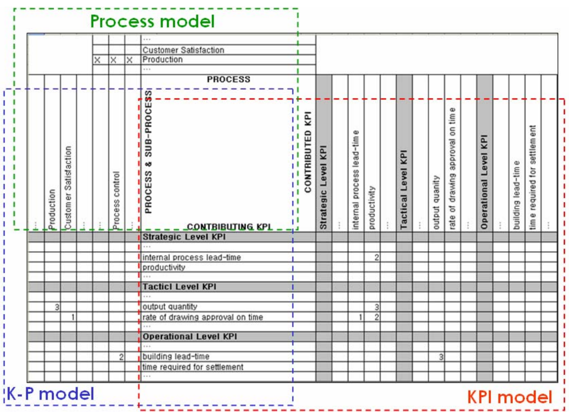

Figure 6. Integrated view of PPMF (simplified)

3. Two-stage process analysis

The process analysis is divided into two stages: macro analysis and micro analysis. All process performance must be related to the enterprise strategy. Therefore, the most influencing business processes on specific SLK are selected at the macro analysis stage. Afterwards, the performance of selected as-is process is reviewed, and to-be process is newly designed. The performance prediction of new process is evaluated by using process simulation at the micro analysis stage.

3.1. Macro process analysis

What-if analysis is executed during the macro analysis stage to tackle the problem of which processes should be executed successfully to attain the goal of a specific SLK. First of all, we find the TLK which has the biggest contribution value to the specific SLK in the KPI model, and then find business process that directly influences the TLK in the K-P model. If there is no process that has a direct influence on the TLK, we alternatively find an OLK that has the biggest contribution value to the TLK in the KPI model, and then identify a sub-process that influences the selected OLK in the K-P model. The higher level process name of the specific sub-process can be found in the process model.

We can also analyze the problem of which SLK is influenced if the execution of a specific process or sub-process is improved. In this case, we find the TLK that is influenced by the specific process in the K-P model, and then determine the SLK that is contributed by the selected TLK in the KPI model. At the sub-process level, we also find the OLK that is influenced by the specific sub-process in the K-P model, and then find the TLK that is contributed by the selected OLK. Finally, we determine the SLK that is affected by the selected TLK in the KPI model.

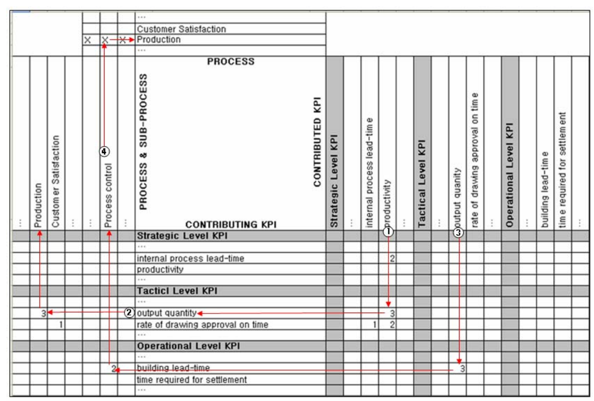

Figure 7. What-if analysis based on PPMF of S company

In the case of S company, the problem remains of which process and sub-process we have to improve for enhancing the ‘productivity’ SLK. By using the proposed PPMF, we identify that the major contributing TLK to the ‘productivity’ SLK is ‘output quantity’ in the KPI model (number ① arrow), as shown in figure 7. Then we determine that the major influencing process on ‘output quantity’ TLK is ‘production’ process in the K-P model (number ② arrow). To find the major influencing sub-process, we see that the major contributing OLK to the ‘output quantity’ TLK is ‘building lead-time’ in the KPI model (number ③ arrow). This result indicates that the major influencing sub-process on ‘building lead-time’ OLK is ‘Process control’ in the K-P model. Finally, we determine that ‘Process control’ is the sub-process of ‘Production’ process in the process model (number ④ arrow).

3.2. Micro process analysis

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At the micro analysis stage, the performance measures of current as-is process are reviewed, and the process simulation for performance prediction of newly designed process is executed. For example, we found the results of macro analysis at the case company S as follows: Target SLK is 'cost'. Major contributing TLK to 'cost' SLK is 'inventory cost'. Major influencing process upon 'inventory cost' TLK is 'purchasing' process and major influencing sub-process to 'purchasing' process is 'placing purchase order'. Therefore, 'placing purchase order' sub-process is selected to enhance the 'cost' SLK. After the review of current performances of 'placing purchase order' sub-process, to-be process is redesigned as shown in figure 8.

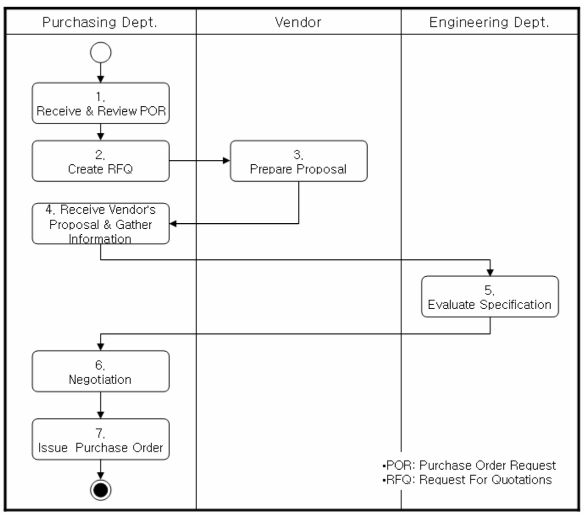

Figure 8. To-be process of placing purchase order

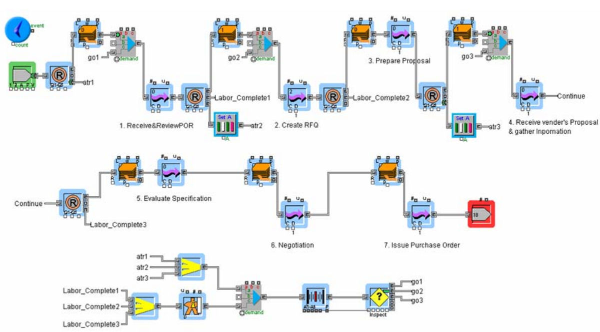

Figure 9. Simulation model using Extend

For performance prediction of new process, simulation model is built. Figure 9 shows the simulation model using Extend [17]. The processing time data for each activity for placing purchase order is shown on table 1.

Table 1. Processing data for each activity

Activity	Time (Normal Distribution)		Labor Pool
	Average (Hr)	Variation (Hr)	
1.Receive &Review POR	3	0.5	Purchasing Dept.
2.Create RFQ	8	1	Purchasing Dept.
3.Prepare proposal	8	2	N/A
4.Receive vendor's Proposal & gather Information	1	0.1	Purchasing Dept.
5.Evaluate Specification	4	1	N/A
6.Negotiation	4	2	Purchasing Dept.
7.Issue Purchase Order	2	0.2	Purchasing Dept.

By using the result of simulation execution for one month simulation period (160 hours), various performance metrics are measured. Average throughput time for each purchase order request (POR) from various departments in the new process is 46.4 hours. POR throughput time trend is shown in figure 10.

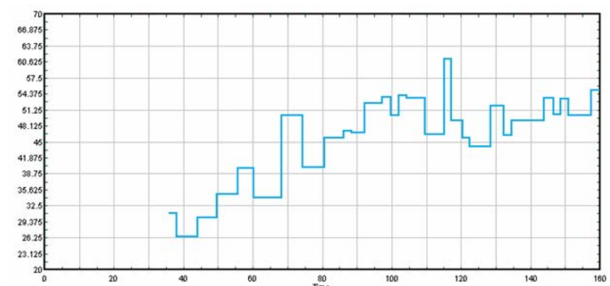

Figure 10. POR throughput time trend

Activity 1, 2, and 4 in figure 8 is executed by labor pool comprised of 4 workers. And each activity 6 and 7 has a dedicated worker. Average resource utilization of labor pool is 88 %. Labor pool resource utilization trend is shown in figure 11.

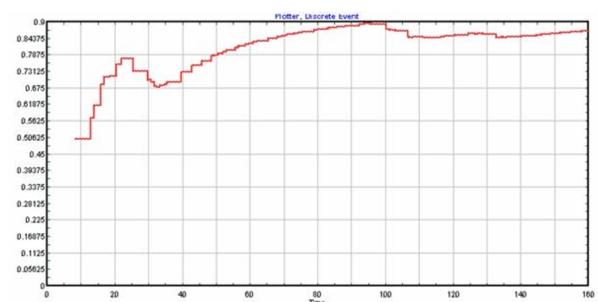

Figure 11. Labor pool resource utilization trend

As explained above using the example of case company S, we can predict the performance of newly designed to-be process and compare it with that of as-is process easily at the micro analysis stage.

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4. Conclusions

A well-designed process analysis enables us to evaluate business process through what-if analysis and to improve process design. This paper has proposed a two-stage process analysis comprised of macro and micro analysis. For structuring enterprise performances aligning with business processes, process-based performance measurement framework (PPMF) is also presented.

By using the proposed two-stage process analysis, company practitioners involved in the process innovation projects can easily and efficiently find the most influencing processes upon enterprise strategy, and evaluate the performance prediction of newly designed process. In other words, the most influencing business processes on specific strategic level KPI are selected at the macro analysis stage by using PPMF. Afterwards, the performance of selected as-is process is reviewed, and to-be process is newly designed. The performance prediction of new process is conducted by using process simulation at the micro analysis stage.

The implementation of current PPMF is a stand-alone system since it is built on an Excel sheet. And it is not integrated with commercial simulation tool. Therefore, as a further research, it is required to develop integrated business process analysis (BPA) tool to support PPMF.

5. Acknowledgements

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